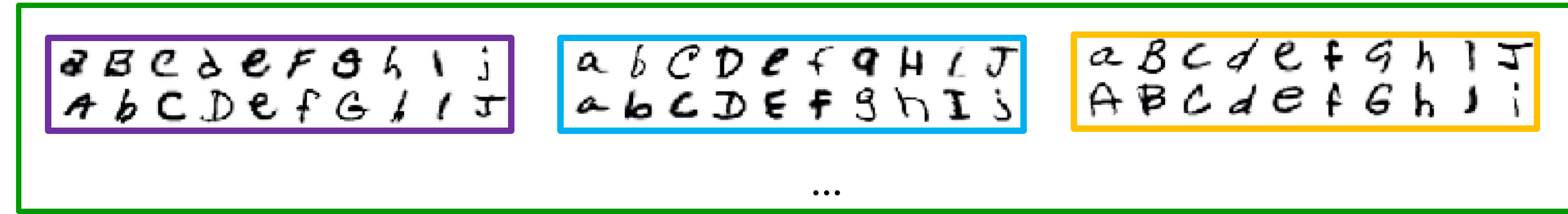


# Promoting Vector Sparsity with Proximal L1 Regularization in Federated Learning for Heterogeneous Datasets

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## Background

### Motivation for Federated Learning



1 Client → Not Enough Data      Many Clients → Enough Combined Data

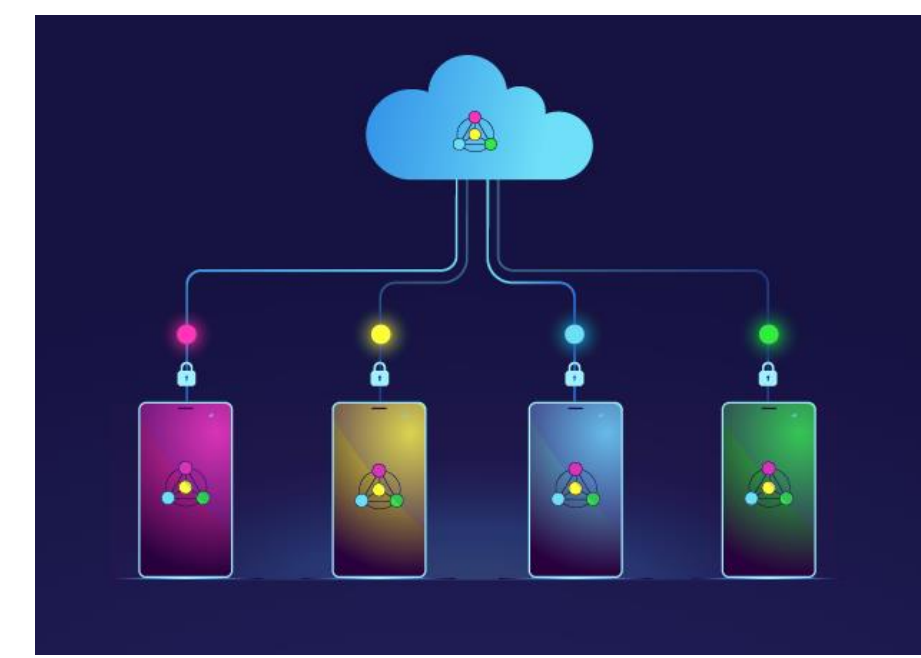
In fields like healthcare and personalized AI solutions like ChatGPT, one individual's data is not enough to train a model.

**Approach:** combine clients' data to have sufficient combined data

**Concern:** must preserve privacy of each client in the process

### Federated Learning – Existing Framework

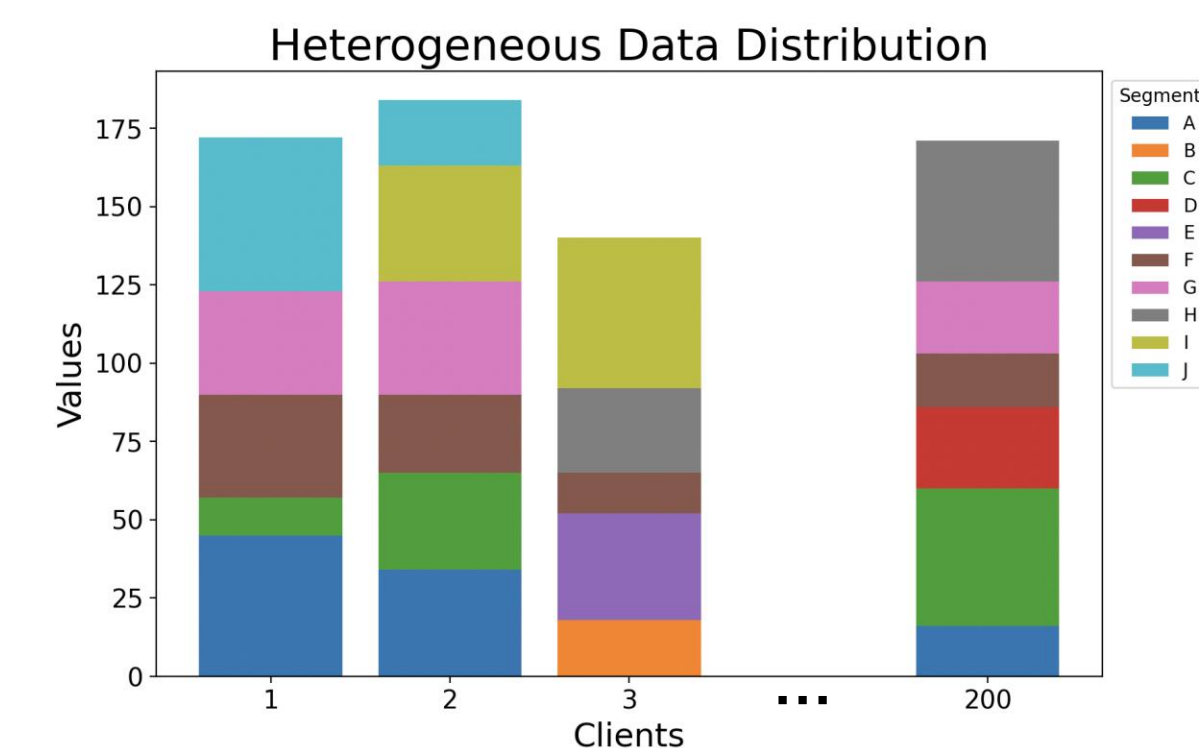
1. Centralized server that receives all  $n$  clients' data
2. Runs stochastic gradient descent on each client's individual dataset to get  $n$  solution vectors
3. Average the  $n$  solutions to get a global solution
4. Pass this global solution vector to clients
5. Repeat steps 2-4 with new solution vector



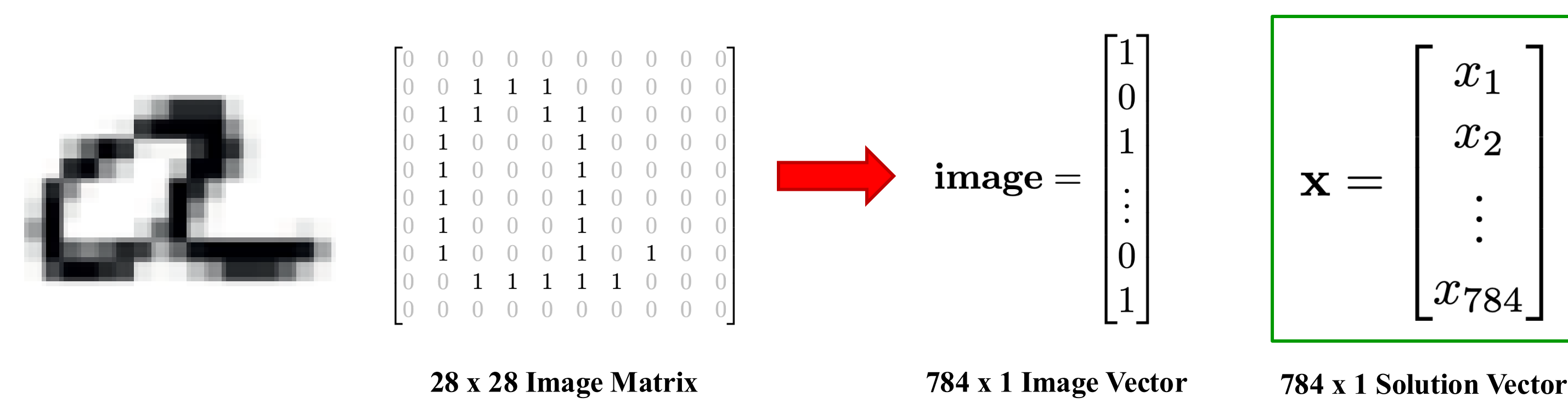
★ This approach enables **continual model updates**: if we have new client data, we can update the model by sending it to the server, instead of restarting the training process with the new dataset. It also **preserves individual client privacy** by using a server that has access only to the solution vector, not the actual data

### Remaining challenges

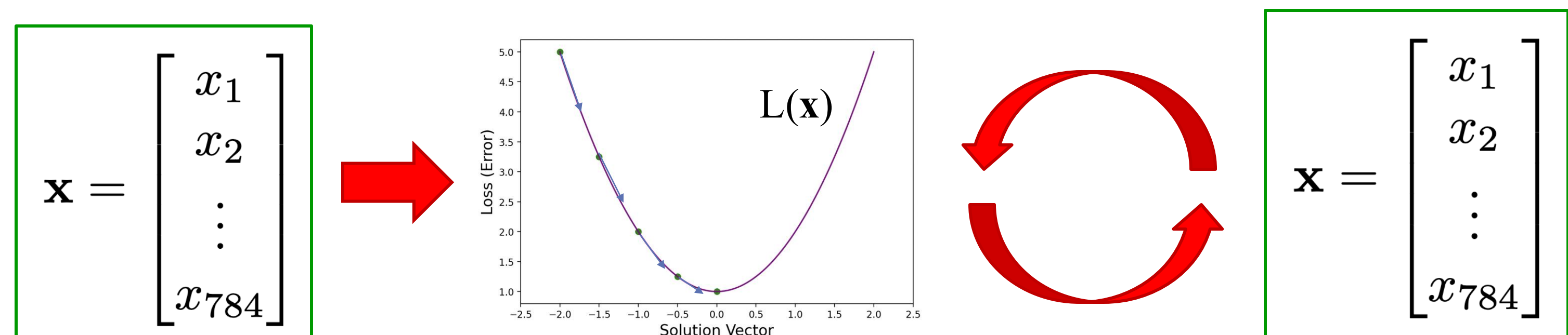
1. **Heterogeneity** – each client can have drastically different; global solution works well for some clients and poorly for others
2. **High Dimensional Solution**– for complex applications, the solution vector will have lots of nonzero values, each one requiring storage space



### Task Background: Image Classification



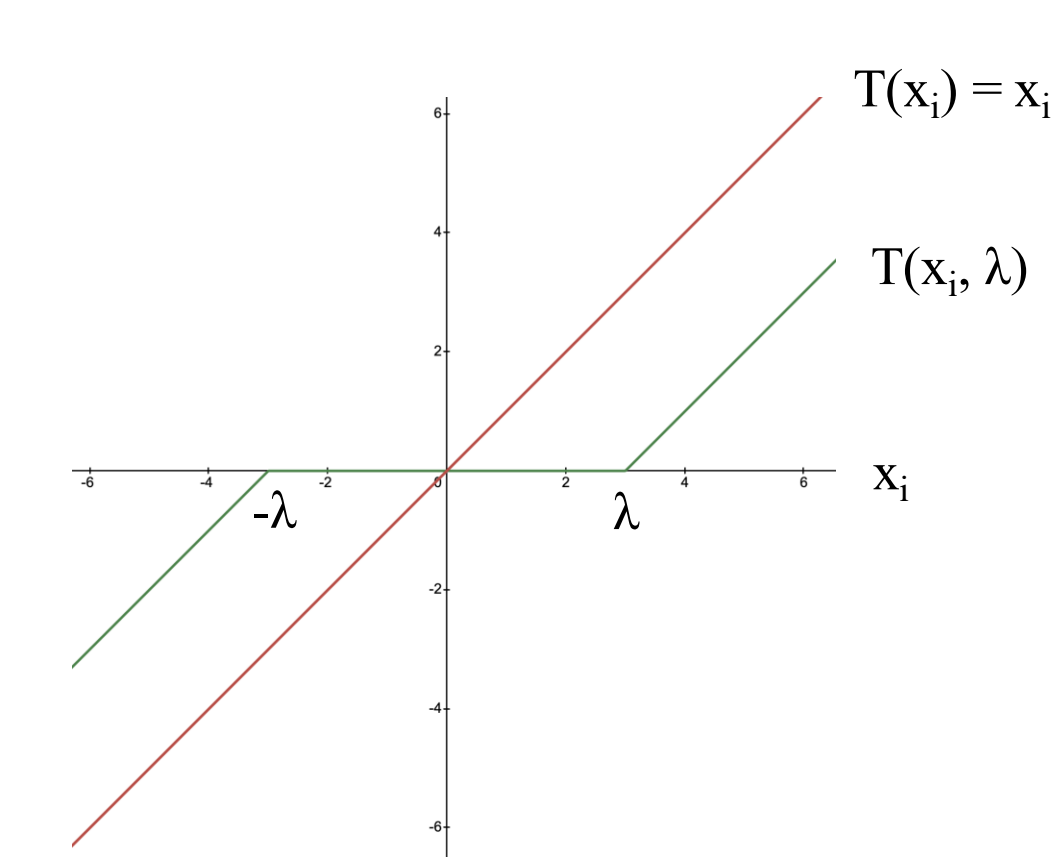
$\text{image} \cdot \mathbf{x} = \text{output}$  (some number)      Classification =  $\begin{cases} a & \text{if output} > \text{threshold} \\ \text{not } a & \text{otherwise} \end{cases}$



Guess Solution Vector      See how well Solution Vector performs with Loss Function      Update Solution Vector based on gradient descent

## Experiments on Promoting Sparsity

### Prox of L1 Norm



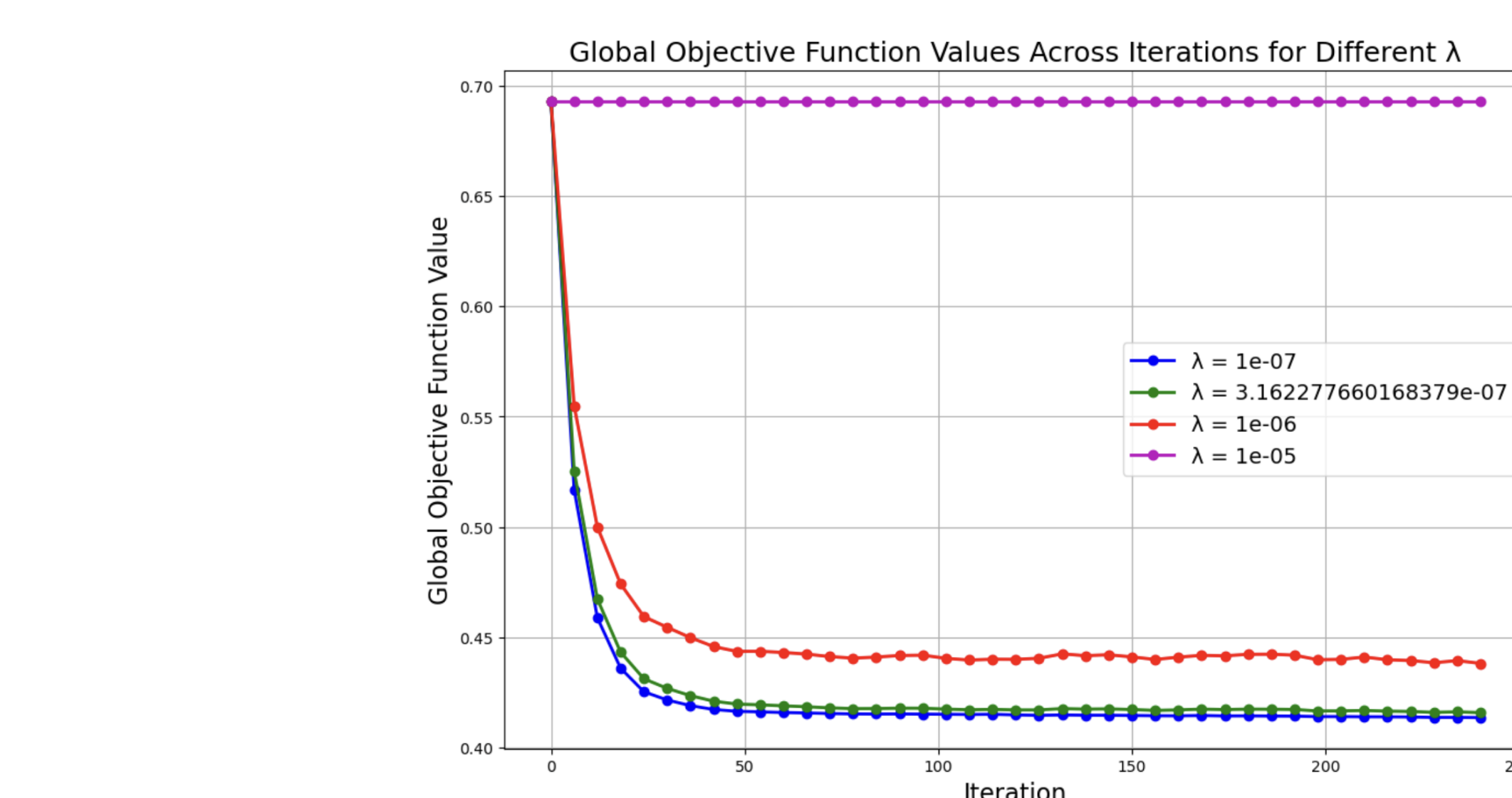
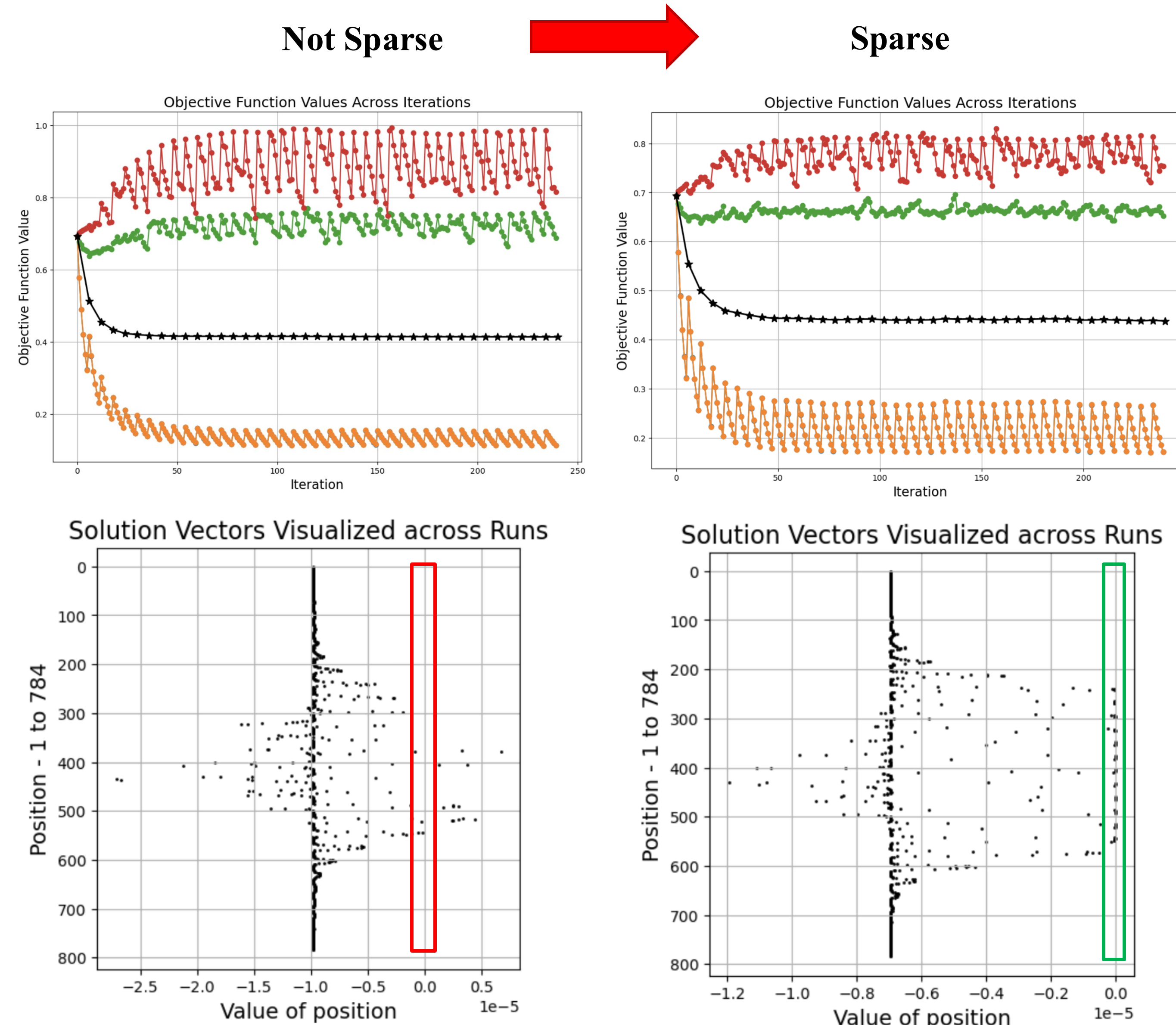
$$T(x_i, \lambda) = \begin{cases} x_i - \lambda, & \text{if } x_i \geq \lambda \\ x_i + \lambda, & \text{if } x_i \leq -\lambda \\ 0, & \text{otherwise} \end{cases}$$

### Intuition:

Values in the solution vector that are:

- close to zero → set to zero
- far away from zero → brought closer to zero

★ This approach promotes vector sparsity by giving us a solution vector that requires less storage space since it has more zeros



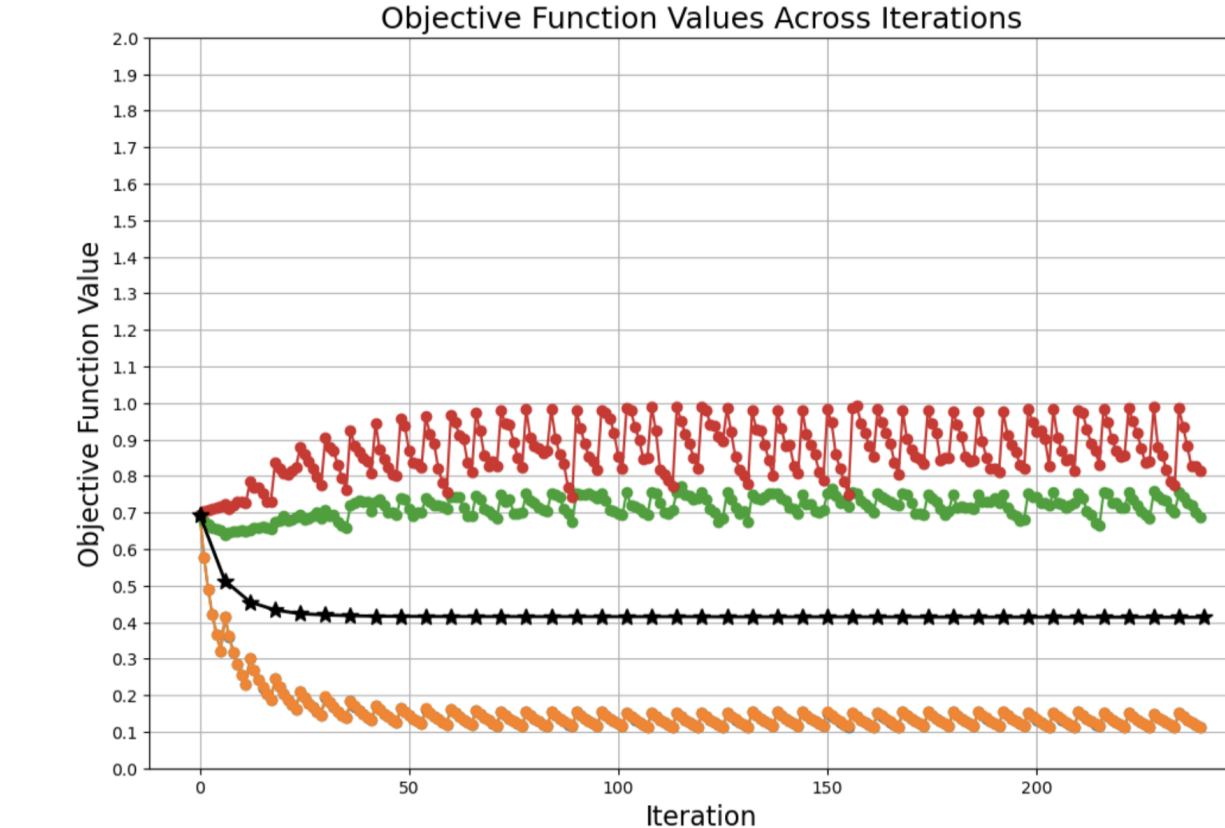
### Hyperparameter Tuning:

Determining an optimal  $\lambda$

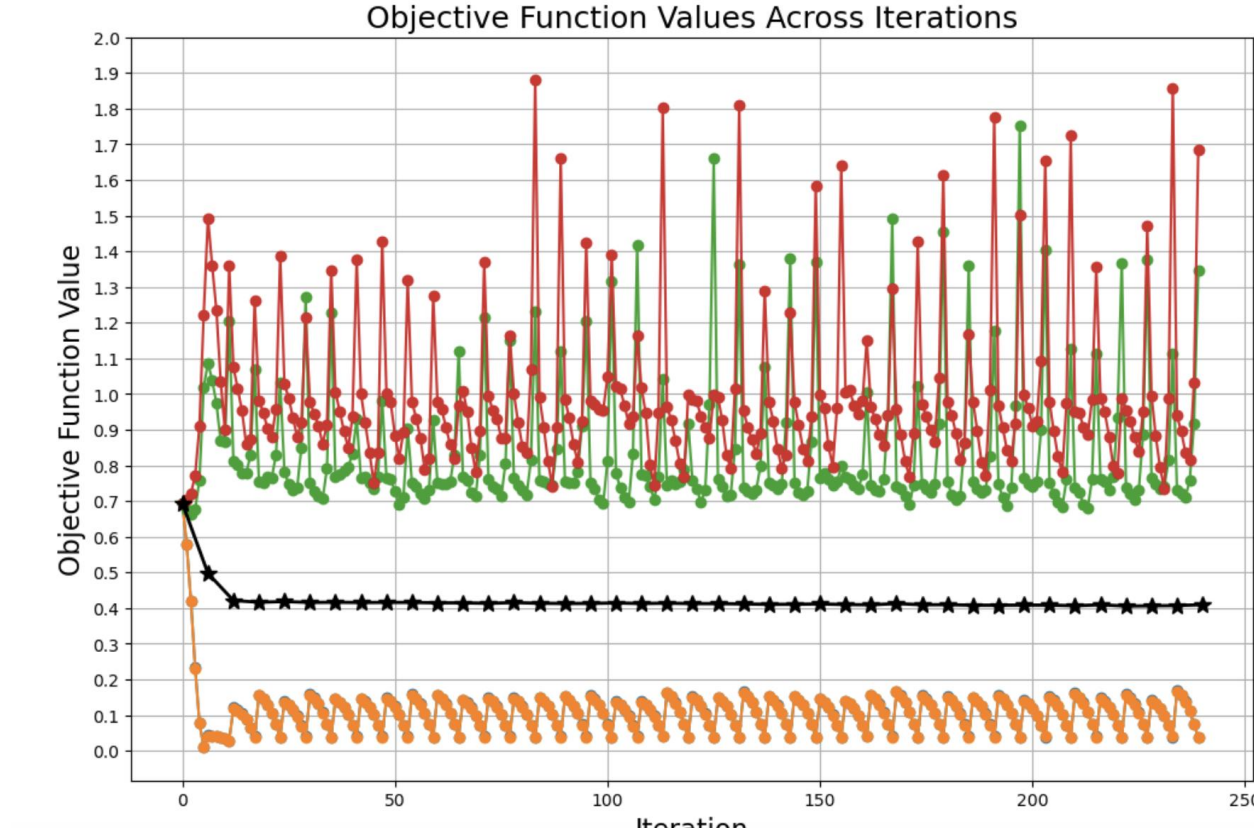
- $\lambda$  too big → all the values are forced to be zero at each communication round; no improvement in loss function throughout learning process
- Smaller  $\lambda$  → improves global loss

## Experiments on Addressing Heterogeneity

### Federated Averaging

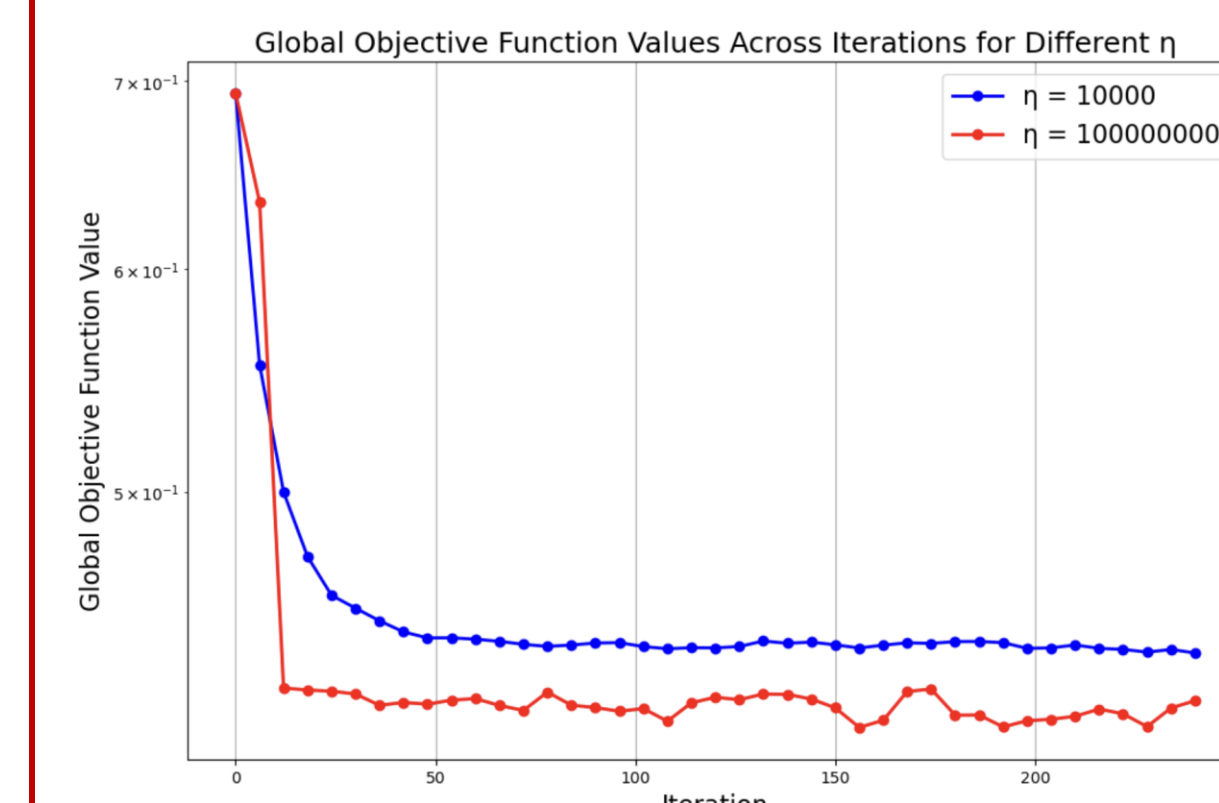


### FedProx



**FedProx:** addresses heterogeneity by incorporating a term in the loss function that penalizes clients for moving away from the global solution

- Individual clients may have more fluctuations
- Their local solutions are closer to the global solution → better to generalize on unseen data



$$L_{FP}(\mathbf{x}, \mathbf{x}_{global}) = L(\mathbf{x}) + \frac{\eta}{2} \|\mathbf{x} - \mathbf{x}_{global}\|^2$$

$F(\mathbf{x}) = L(\mathbf{x}) + \eta * (\text{deviation from global model})$

### Hyperparameter Tuning:

- Large  $\eta$  → improves global model
- Small  $\eta$  → negligible effect on training process

## Conclusion

Based on our experiments, introducing the L1 norm prox operator at each communication round in the global step helped to promote vector sparsity while maintaining robust model performance. Combining this modification with the FedProx method also resulted in an incremental improvement in overall global loss. We also observed the sensitivity of model performance to the exact values of the hyperparameters associated with these approaches. We conclude that we can leverage these techniques and considerations to improve model performance and generalization in heterogeneous settings.

## Future Direction

Federated learning has a lot of potential application in fields off individual data is insufficient to train a model, but collective data is. Also, this framework is especially useful if we want to build off a current model with additional data. We can extrapolate the results on the MNIST dataset in our experiments to other applications. For example, if we want to improve an LLM using recent conversations it has had with users, we can leverage federated learning to accomplish that while also preserving individual privacy. We can also use federated learning to train models on patient healthcare data. This can discover potentially life-saving insights, while preserving individual patient privacy.



## Acknowledgments

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